

MATH 345 Skeleton Notes

Unit 6: Constraints and SVD

Section 6.5: Low-rank approximation and compression

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Application question. How can a matrix be approximated by keeping only its strongest directions?

The SVD writes a matrix map as a sum of directional pieces. If

$$A = U\Sigma V^T$$

has rank r , then

$$A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T.$$

The matrix

$$\sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

has rank one. It measures the input coordinate in the direction $\mathbf{v}_i$, stretches by σ_i , and sends the result in the output direction $\mathbf{u}_i$.

Definition: Rank- k reconstruction Let

$$A = U\Sigma V^T$$

be an SVD of a rank- r matrix A . For $1 \leq k \leq r$, the rank- k reconstruction of A is

$$A_k = U_k \Sigma_k V_k^T = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T,$$

where U_k contains the first k columns of U , Σ_k contains the first k singular values, and V_k contains the first k columns of V .

The matrix A_k keeps the k largest singular directions and discards the rest. Its rank is at most k .

Activity: Rank-one reconstruction of a diagonal stretch

Return to the diagonal stretch from Section 6.3:

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

For this matrix, the right and left singular vectors are

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and the singular values are

$$\sigma_1 = 3, \quad \sigma_2 = 1.$$

1. Write A as

$$3\mathbf{e}_1\mathbf{e}_1^T + \mathbf{e}_2\mathbf{e}_2^T.$$

2. Compute the rank-one reconstruction

$$A_1 = 3\mathbf{e}_1\mathbf{e}_1^T.$$

3. For

$$\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix},$$

compute $A\mathbf{x}$ and $A_1\mathbf{x}$.

4. Which input direction is discarded by A_1 ?

5. What is $\text{rank}(A_1)$?

We have

$$\begin{aligned}\mathbf{e}_1 \mathbf{e}_1^T &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}$$

and

$$\begin{aligned}\mathbf{e}_2 \mathbf{e}_2^T &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.\end{aligned}$$

Thus

$$\begin{aligned}
3\mathbf{e}_1\mathbf{e}_1^T + \mathbf{e}_2\mathbf{e}_2^T &= 3\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = A.
\end{aligned}$$

The rank-one reconstruction is

$$A_1 = 3\mathbf{e}_1\mathbf{e}_1^T = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}.$$

Therefore

$$\begin{aligned}
A \begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 3a \\ b \end{bmatrix}, \\
A_1 \begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 3a \\ 0 \end{bmatrix}.
\end{aligned}$$

The rank-one reconstruction keeps the e_1 direction and discards the e_2 direction. The rank of A_1 is 1.

Note: Energy retained A common way to summarize how much of a matrix has been kept is to use squared singular values. The energy retained by the first k singular values is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}.$$

This number is between 0 and 1. Larger values mean that the rank- k reconstruction keeps more of the matrix's dominant directions.

Activity: Choosing k from singular values

Suppose the nonzero singular values of a matrix are

10, 4, 1, 0.5, 0.1.

1. Compute the total squared singular-value energy.
2. Compute the energy retained by $k = 1$.
3. Compute the energy retained by $k = 2$.
4. What is the smallest k that retains at least 90% of the energy?

The squared singular values are

$$100, 16, 1, 0.25, 0.01.$$

The total squared energy is

$$100 + 16 + 1 + 0.25 + 0.01 = 117.26.$$

For $k = 1$, the retained energy is

$$\frac{100}{117.26} \approx 0.853.$$

For $k = 2$, the retained energy is

$$\frac{100 + 16}{117.26} = \frac{116}{117.26} \approx 0.989.$$

Thus $k = 2$ is the smallest value that retains at least 90% of the energy.

Note: Rank- k reconstruction in NumPy In NumPy, the singular values are returned as a one-dimensional array s . The diagonal matrix Σ_k is built from $s[:k]$.

```
U, s, Vt = np.linalg.svd(A, full_matrices=False)
Ak = U[:, :k] @ np.diag(s[:k]) @ Vt[:k, :]
```

The matrix $U[:, :k]$ keeps the first k left singular vectors. The matrix

$Vt[:k, :]$ keeps the first k rows of V^T , which correspond to the first k right singular vectors. The product A_k is the rank- k reconstruction. For a quick reference on SVD commands, diagonal matrices, slices, and matrix multiplication, see the programming appendix sections NumPy Linear Algebra Commands, Constructing Common Arrays and Design Matrices, Indexing and Slicing, and Elementwise Arithmetic Versus Linear Algebra.

Activity: Reading rank- k reconstruction code

The following code computes a rank-one reconstruction of the diagonal stretch.

```
import numpy as np

A = np.array([[3., 0.],
              [0., 1.]])

U, s, Vt = np.linalg.svd(A, full_matrices=False)

k = 1
Ak = U[:, :k] @ np.diag(s[:k]) @ Vt[:k, :]

s, Ak
```

Output:

```
(array([3., 1.]),
 array([[3., 0.],
        [0., 0.]])
```

1. What are the singular values of A ?
2. Which singular value is kept when $k = 1$?
3. What is the matrix A_1 ?

4. Which input direction does A_1 discard?
5. Why is $Vt[:k, :]$ used instead of $Vt[:, :k]$?

The singular values are 3 and 1. When $k = 1$, the reconstruction keeps only the singular value 3. The reconstructed matrix is

$$A_1 = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}.$$

It discards the second input direction. The slice $Vt[:k, :]$ keeps the first k rows of V^T , which are the transposed right singular vectors used in the reconstruction. The slice $Vt[:, :k]$ would keep columns of V^T , which is the wrong shape and the wrong mathematical object for this product.

Activity: Redundant features revisited by singular values

Return to the redundant-feature matrix from Unit 2:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

The third column is the sum of the first two columns.

The following code computes the singular values and the last row of V^T .

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

X = np.array([[1., 2., 3.],
              [2., 4., 6.],
              [0., 1., 1.],
              [1., -1., 0.]])

U, s, Vt = np.linalg.svd(X, full_matrices=False)

s, Vt[-1, :]
```

Output:

```
(array([8.478, 1.459, 0.   ]),
 array([-0.577, -0.577, 0.577]))
```

1. What does the zero singular value say about the rank of X ?
2. What null-space direction does the last row of V^T point along?
3. How does this match the Unit 2 relation among the columns?
4. If $X\mathbf{c} = \mathbf{y}$, what happens to $X(\mathbf{c} + t\mathbf{z})$, where

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} ?$$

The zero singular value says that the three columns of X contain only two independent directions, so

$$\text{rank}(X) = 2.$$

The last row of V^T is approximately

$$[-0.577 \quad -0.577 \quad 0.577],$$

which points in the same direction as

$$-\frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

This matches the Unit 2 relation

$$\text{column 1} + \text{column 2} - \text{column 3} = \mathbf{0},$$

or

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

Therefore, if $X\mathbf{c} = \mathbf{y}$, then

$$\begin{aligned} X(\mathbf{c} + t\mathbf{z}) \\ &= X\mathbf{c} + tX\mathbf{z} \\ &= \mathbf{y}. \end{aligned}$$

The null direction gives different coefficient vectors with the same output.

Note: Low-rank updates A low-rank update changes a large matrix by adding a product

$$BC,$$

where

$$B \in \mathbb{R}^{m \times r}, \quad C \in \mathbb{R}^{r \times n},$$

and r is small. Since

$$\text{rank}(BC) \leq r,$$

the update BC changes the matrix through at most r independent directions.

This is useful when a full $m \times n$ matrix would be expensive to store or change. The matrix BC can be stored using

$$mr + rn$$

numbers instead of mn numbers.

In model-adaptation settings, W may be a large existing weight matrix and BC may represent a small trainable update. The linear-algebra point is independent of that application: the update BC has rank at most r and can be stored using many fewer numbers than a full $m \times n$ matrix.

Activity: Rank of an update

Let

$$W \in \mathbb{R}^{1000 \times 800}, \quad B \in \mathbb{R}^{1000 \times 4}, \quad C \in \mathbb{R}^{4 \times 800}.$$

Define

$$W_{\text{new}} = W + BC.$$

1. What is the shape of BC ?
2. What is the largest possible rank of BC ?
3. How many numbers are needed to store W ?
4. How many numbers are needed to store B and C ?
5. Why is BC called a low-rank update?

The product BC has shape

$$1000 \times 800,$$

so it can be added to W .

The rank of a product is at most the inner dimension, so

$$\text{rank}(BC) \leq 4.$$

The full matrix W stores

$$1000 \cdot 800 = 800000$$

numbers. The two update matrices store

$$1000 \cdot 4 + 4 \cdot 800 = 4000 + 3200 = 7200$$

numbers.

The update is called low-rank because the change BC has rank at most 4, even though it has the same shape as W .

Warning: Low-rank update versus low-rank matrix A low-rank

update does not mean that the whole matrix

$$W_{\text{new}} = W + BC$$

has low rank. The update BC has rank at most r . The matrix $W + BC$ may still have large rank.